

THOMAS CYRIAC

Smart Contract Engineer • DeFi Protocol Development & Security

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PROFILE

Smart contract engineer with 6 years of experience, writing production Solidity since 2021. Lead contract engineer on BankX (2021–2026), a commodity-pegged stablecoin protocol live across 8 EVM chains, audited by CoinFabrik and covered by an Immunefi bug bounty. Also builds the systems around protocols — AI treasury automation, a peg-correction CLI and self-hosted AI agent infrastructure — and contributes Rust to OpenZeppelin and Nethermind. Focused on protocol security: reproducing bounty findings in Foundry and debugging production failures on live chains.

TECHNICAL SKILLS

Languages	Solidity, Rust, TypeScript / JavaScript, Python, Java
Contracts	ERC-20 / BEP-20, staking & liquidity pools, bonding curves, on-chain governance, upgradeable proxies & factories, AMM and lending integrations
Chains	Ethereum, BNB Chain, Arbitrum, Optimism, Avalanche, Base, Polygon, PulseChain, Stellar (Soroban)
Tooling	Hardhat, Foundry (forge / cast), Truffle, Ganache, OpenZeppelin, Chainlink Price Feeds & Automation, mainnet-fork testing, Ethers.js, Node.js
Security	Foundry proof-of-concept reproduction, Immunefi submission triage, reentrancy & flash-loan attack patterns, access-control review, oracle and peg failure modes
Adjacent	Node.js, MongoDB, Docker, Caddy, DigitalOcean, Claude API, MCP agent orchestration

EXPERIENCE

AI Infrastructure Engineer — LockBox (AI agent platforms) Apr 2026 – Present • Remote

- Deployed and hardened the self-hosted agent-orchestration stack that drives **AgentXSD**, an autonomous system building a multi-asset, multi-chain payment stack; the client now runs it independently.
- Hardened the deployment (Docker Compose, Caddy, MongoDB, IPv4-only networking) and added run-control gating so autonomous agents could not push unreviewed changes.

Lead Smart Contract Engineer — BankX Protocol (direct client contract) Oct 2021 – Aug 2026 • Remote

- Wrote and deployed the Solidity suite behind **XSD**, a stablecoin pegged to one gram of silver through Chainlink XAG/USD price feeds, live on **8 EVM chains**.
- Implemented Sybil-resistant minting: a per-wallet mint record with weighted-average interest that binds redemption to the minting wallet, closing a two-wallet mint/redeem attack.
- Wrote and ran Foundry tests to reproduce and validate external **Immunefi** researcher submissions before remediation.
- Debugged production failures: a Chainlink Automation gas-limit bug (diagnosed with a Chainlink Labs engineer), and *Price Manipulation detected* reverts on Polygon when XSD spot price drifted from the PID controller's moving average, fixed by recalibrating the deviation threshold.
- Built the protocol's AI-driven autonomous treasury modules and a Node.js CLI that reads live pool reserves and computes the exact XSD to add or remove to restore the peg.

Smart Contract Developer — Freelance • Upwork (Top Rated Plus, 100% Job Success) Apr 2021 – Present • Remote

- Based Agents** — built the full contract suite (Protocol Rewards, Bonding Curve, Bag Token, Bag Governance, upgradeable Bag Factory with split implementation and factory addresses); Hardhat tests against a mainnet fork covering governance amendments and market-graduation paths.
- Stellar Soroban (Rust)** — fixed bugs in a client's Soroban codebase, deployed it to Stellar testnet and Futurenet, and trained their team on deployment and operation.
- Built a staking and liquidity contract combining **PancakeSwap** AMM swaps with **Venus Protocol** lending on BNB Chain.
- Contributed to a decentralized perpetual futures platform; built an IPFS-backed NFT marketplace with the collection listed on OpenSea.

Blockchain Developer — Nord Finance Jul 2021 – Sep 2021 • Remote

- Built automated test suites (Hardhat, Ganache) for the DeFi contracts behind interest-bearing savings products — deposits, withdrawals, approvals, yield accounting and access control — run against local and mainnet-forked state.
- Contributed to the design of a decentralized loan application and its user flows.

Earlier 2020 – 2024

- Machine learning & deep learning** (2020–21) — GIEOM, Appciliious, LAMBDA VISION: face alignment for KYC (dlib, OpenCV), a U-2-Net background-removal service on AWS, TensorFlow 2 research.
- Network Engineer, Ctrlbit** (2024) — SD-WAN product on MikroTik hardware; dual-SIM failover with GenieACS / TR-069 and OpenWRT.

OPEN SOURCE

- OpenZeppelin/soroban-helpers **#121, #122** — **merged**. Separated build and simulate in TransactionBuilder with a standalone `simulate_transaction()`; added Extend Footprint TTL support to keep Soroban ledger entries from being archived.
- NethermindEth/stellar-private-payments **#200** — **merged**. Embedded the Groth16 verification key into the verifier WASM at compile time, removing a storage write at deploy and a read on every verify.
- smartcontractkit/ccip-tools-ts **#124** — **open**. Made manual execution loop over several failed cross-chain messages in one session instead of one run per message.

EDUCATION & CERTIFICATION

B.Tech Computer Science — SRM Institute of Science & Technology, 2017–2021 • CGPA 7.8 | **MTCNA** certified, Oct 2024